Diagnosing Slow Windows 11 Performance

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1. Problem Description

A Windows 11 workstation was experiencing noticeably slow performance during normal use, productivity and gaming. Symptoms included long boot times, delayed application launches, and overall system lag when multitasking. The issue impacted productivity and required investigation to identify potential hardware, software, or configuration-related causes.

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2. System Environment

- **Operating System:** Windows 11 Pro (64-bit)
- **Processor:** Ryzen 5 5600x
- **Memory:** 16 GB RAM DDR4
- **Storage:** 1TB GB SSD
- **Graphics Card:** RX 6600
- **Use Case:** General productivity, with some gaming.

3. Troubleshooting Approach

A structured troubleshooting approach was used, beginning with software and configuration checks before moving to system integrity and driver validation. Tools included Task Manager, Windows Security, Command Prompt, and Windows Update.

4. Troubleshooting Steps

4.1 Startup and Background Applications

- Opened **Task Manager → Startup**
- Identified non-essential applications launching at boot
- Disabled unnecessary startup items to reduce background CPU and memory usage

4.2 Resource Utilization Analysis

- Monitored system performance using **Task Manager → Performance**
- Observed high memory utilization during normal multitasking, indicating limited available RAM

4.3 Storage Health and Cleanup

- Verified available disk space and found the SSD was over 80% utilized
- Ran **Disk Cleanup** to remove temporary files and system cache

4.4 System File Integrity Check (Command Prompt)

- Opened **Command Prompt as Administrator**
- Executed the following command to scan and repair corrupted system files:

  sfc /scannow DISM /Online /Cleanup-Image /RestoreHealth

- The scan detected and repaired corrupted Windows system files that could contribute to performance issues

4.5 Malware and Security Scan

- Performed a **full system scan** using Windows Defender
- No malware or unwanted software was detected

4.6 Driver and Graphics Update Verification (AMD)

- Checked **Device Manager** for display adapter status
- Verified AMD graphics drivers using **AMD Adrenalin Software**
- Confirmed drivers were up to date, ensuring optimal system and graphics performance

4.7 Windows

- Installed all pending **Windows Updates**

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5. Findings

The primary contributors to the slow performance were:

- Excessive startup applications consuming system resources
- High memory usage due to limited RAM
- Insufficient free disk space impacting system responsiveness

No hardware failure or malware-related issues were identified.

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6. Resolution

After disabling unnecessary startup applications, repairing system files, cleaning up storage, and confirming up-to-date AMD drivers, system performance improved significantly. Boot times decreased, applications launched faster, and overall responsiveness was restored.

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7. Recommendations

- Upgrade system memory from **16 GB to 32 GB** for improved multitasking performance
- Periodically review startup applications and installed software
- Maintain at least **20–25% free disk space** on the primary drive
- Continue regular system updates and security scans

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8. Outcome

The system returned to stable and responsive performance following corrective actions. This troubleshooting exercise demonstrates the application of practical IT support skills, including system diagnostics, command-line tools, driver management, and preventative maintenance.